

PROJECT REPORT

Incident Lifecycle Automation in ServiceNow

ServiceNow IT Incident Management Automation

Project Type	Individual Project
Platform	ServiceNow
Project Area	IT Service Management (ITSM)
Core Module	Incident Management
Date	19 September 2026

Report Structure

This report follows the supplied Final Report Template: Introduction, Ideation Phase, Requirement Analysis, Project Design, Project Planning & Scheduling, Functional and Performance Testing, Results, Advantages & Disadvantages, Conclusion, Future Scope, and Appendix. ■filecite■turn2file0■L2-L30■

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1. INTRODUCTION

1.1 Project Overview

Incident Lifecycle Automation in ServiceNow is an individual IT Service Management project that organizes and automates the handling of IT incidents. The solution uses ServiceNow Incident Management to provide a structured lifecycle from incident creation and classification through prioritization, assignment, resolution, closure and reopening. The project also includes supporting knowledge/change information and Platform Analytics dashboards for operational monitoring.

1.2 Purpose

The purpose of the project is to reduce manual handling in incident management, improve consistency in priority calculation and assignment, provide controlled lifecycle processing, and give service desk users and administrators better visibility into incident information.

The project is implemented primarily through ServiceNow configuration and JavaScript-based Client Script logic rather than an external machine-learning model.

2. IDEATION PHASE

2.1 Problem Statement

IT support teams receive incidents with different levels of urgency and business impact. Manual classification, priority selection and routing can lead to inconsistent handling. Incident progress can also be difficult to monitor when lifecycle information is not standardized. The project addresses these problems through a structured ServiceNow incident workflow with automation and analytics.

2.2 Empathy Map Canvas

Perspective	Key Points
Says	I need to report an issue and know what is happening with it.
Thinks	My issue should reach the correct support team with the correct priority.
Does	Creates incidents, provides issue details, checks status and responds to support activity.
Feels	Expects quick handling, transparency and predictable resolution.
Pain Points	Manual classification, unclear ownership, inconsistent priority and limited visibility.
Needs	Simple incident intake, correct routing, lifecycle visibility and useful reporting.

2.3 Brainstorming

Idea	Purpose
Incident Intake	Capture issue details consistently.
Priority Automation	Use Impact and Urgency to calculate priority.
Assignment	Route incidents to Hardware, Software or Network support.
Lifecycle Control	Manage work, resolution, closure and reopening.
Knowledge / Change	Provide supporting related information.
Analytics	Monitor incidents by operational dimensions.

3. REQUIREMENT ANALYSIS

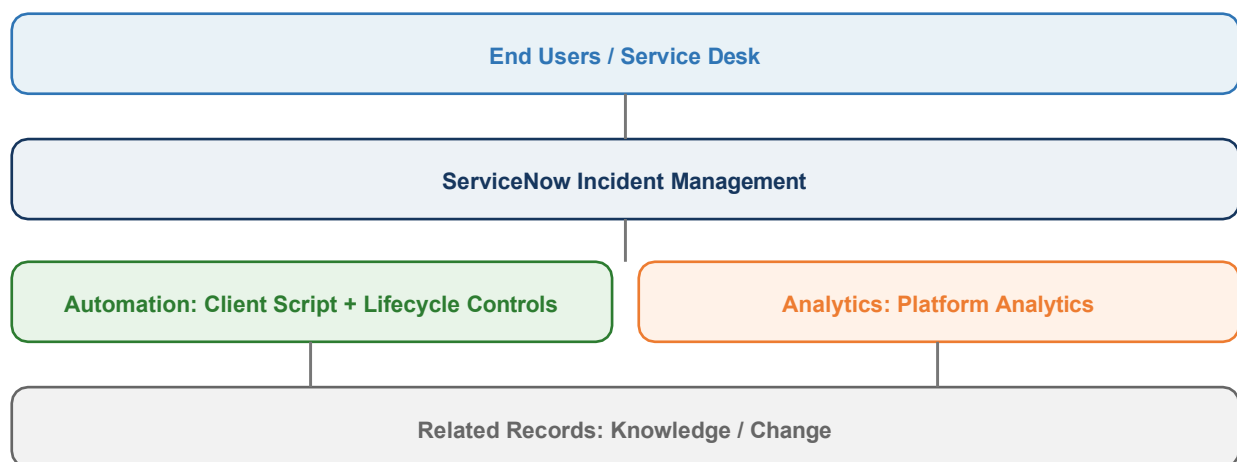
3.1 Customer Journey map

Stage	User Action	System Response	Expected Experience
Report	Create incident	Incident record created	Simple issue submission
Classify	Select category/subcategory	Controlled choices	Consistent classification
Prioritize	Select impact/urgency	Priority calculated	Consistent priority
Assign	Select group/user	Ownership recorded	Clear responsibility
Resolve	Add resolution	Resolution captured	Traceable completion
Close	Complete lifecycle	Incident closed	Clear final status
Follow-up	Reopen if needed	Incident becomes active	Continued support
Monitor	Review dashboard	Analytics update	Operational visibility

3.2 Solution Requirement

ID	Requirement
SR-01	Provide a ServiceNow incident record for IT issue management.
SR-02	Support category and subcategory based incident classification.
SR-03	Automatically calculate priority from Impact and Urgency.
SR-04	Support assignment groups including Hardware, Software and Network.
SR-05	Control incident state and resolution/closure processing.
SR-06	Allow supported reopening of completed incidents.
SR-07	Support related knowledge and change information.
SR-08	Provide Platform Analytics for priority, state, category and assignment group.

3.3 Data Flow Diagram



Incident information enters through the ServiceNow interface and is stored as an incident record. Automation evaluates relevant fields, support groups handle the work, related records provide context, and Platform Analytics uses incident information for monitoring.

3.4 Technology Stack

Component	Use
ServiceNow	Primary ITSM platform and application environment
Incident Management	Core incident lifecycle

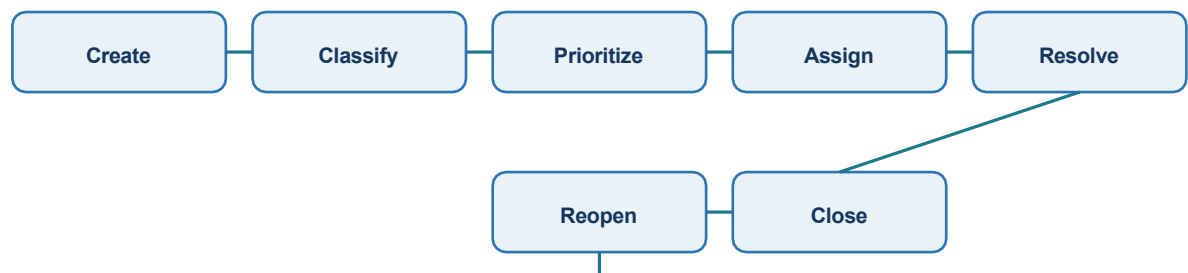
Component	Use
JavaScript	Client-side automation / priority calculation
Client Script	Automatically updates priority from configured fields
App Engine Studio / Scoped Application	Application configuration
Platform Analytics	Dashboards, visualizations and filters
Knowledge Management	Supporting knowledge information
Change Management	Related change activity where required
Web Browser	Access and demonstration

4. PROJECT DESIGN

4.1 Problem Solution Fit

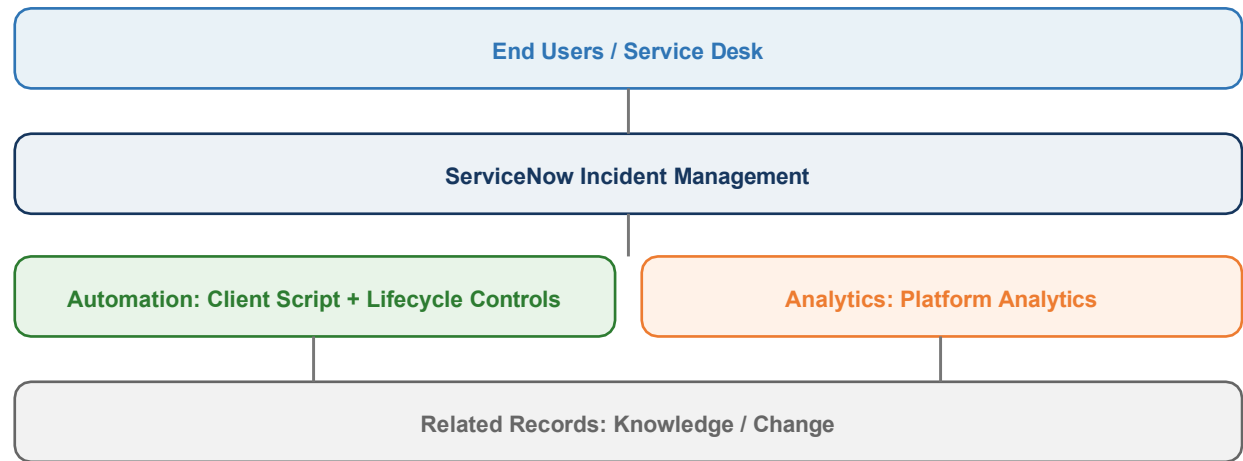
Problem	Designed Solution
Manual priority selection	Impact + Urgency based priority automation
Inconsistent classification	Controlled category/subcategory values
Unclear ownership	Assignment groups and assignees
Incomplete lifecycle tracking	Configured state and resolution handling
Limited supporting information	Knowledge/change related records
Limited operational visibility	Platform Analytics dashboard and filters

4.2 Proposed Solution



The proposed solution standardizes the incident lifecycle. A new incident is created and classified, Impact and Urgency drive priority calculation, the incident is assigned to the appropriate support group, work is performed, resolution information is recorded, and the incident is closed. A supported reopening path allows continued processing when required.

4.3 Solution Architecture



The architecture keeps ServiceNow Incident Management as the central workflow, with Client Script logic supporting priority calculation and Platform Analytics providing operational visibility.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Planned Activities	Output
Ideation	Identify incident-management problem and objective	Problem definition
Requirement Analysis	Identify users, requirements and workflow needs	Requirement specification
Project Design	Design lifecycle, architecture and automation logic	Solution design
Project Planning	Plan configuration, testing and demonstration	Implementation plan
Project Development	Configure ServiceNow, scripts, groups, lifecycle and analytics	Working application
Project Documentation	Prepare project and testing documents	Documentation package
Project Demonstration	Demonstrate incident lifecycle and dashboards	Demonstration evidence

Development Activities

- Configure the ServiceNow application and incident environment.
- Configure incident categories, subcategories and assignment groups.
- Implement and test the Impact/Urgency based Client Script.
- Configure state, resolution and closure behavior.
- Configure knowledge/change related-record handling.
- Build Platform Analytics visualizations and filters.
- Perform functional and performance-oriented testing.
- Prepare final demonstration and documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The supplied Phase 5 reference uses Functional & Performance Testing with FT and PT test-case identifiers.

■ filecite ■ turn1file5 ■ L2-L18 ■ The scenarios below adapt that format to the ServiceNow incident lifecycle.

ID	Scenario	Test Action	Expected Result
FT-01	Incident Form Validation	Create incident with configured fields.	Valid data is accepted and validation works.
FT-02	Priority Automation	Change Impact and Urgency combinations.	Priority updates according to Client Script.
FT-03	Assignment	Select assignment group and assignee.	Incident ownership is recorded correctly.
FT-04	Lifecycle Processing	Move incident through configured states.	Supported state transitions work.
FT-05	Resolution Validation	Provide resolution information and complete.	Resolution is captured and completion is supported.
FT-06	Reopen Processing	Reopen a completed incident.	Incident becomes active again.
PT-01	Form Response Time	Measure incident form loading time.	Loads within acceptable instance response time.
PT-02	Priority Response	Measure Priority update after field changes.	Priority updates promptly.
PT-03	Save Response	Measure incident save/update response.	Record saves without abnormal delay/failure.
PT-04	Dashboard Load	Measure Platform Analytics dashboard loading.	Dashboard loads successfully.
PT-05	Concurrent Access	Use multiple sessions for incident activity.	No abnormal failure or data corruption.
PT-06	Dashboard Concurrency	Open/filter dashboard from multiple sessions.	Dashboard remains usable.
PT-07	Lifecycle Load	Execute multiple incident operations in a controlled test.	System continues processing selected workload.

Performance Measurement Fields

Parameter	Value
Incident form response time	To be recorded during execution
Priority update response time	To be recorded during execution
Incident save/update response time	To be recorded during execution
Dashboard load time	To be recorded during execution
Concurrent users/sessions	To be recorded during execution
Observed errors	To be recorded during execution

No numerical performance measurements are claimed here because the supplied project evidence does not contain measured response-time or load-test values.

7. RESULTS

7.1 Output Screenshots

The following screenshots are taken from the supplied ServiceNow demonstration evidence and document the implemented project output.

1. Remote Access Service and Corporate VPN Service Offering.

dev224508.service-now.com/service_offering.do?sys_id=045cee1a83574f1063f6c8efeeaad3e6&sysparm_record_target=service_offering&sysparm_record_key=045cee1a83574f1063f6c8efeeaad3e6

Offering Corporate VPN

Name	Corporate VPN	Approval group	
Model ID		Support group	
Owned by		Change group	
* Parent	Remote Access	Managed by	
Used for	Production	Version	
Operational status	Operational	Business criticality	2 - somewhat critical
Vendor		SLA	
		Location	
Short Description			
Description			
Comments			

2. Newly created Corporate VPN incident.

dev224508.service-now.com/service_offering.do?sys_id=045cee1a83574f1063f6c8efeeaa3e6&sysparm_record_target=service_offering&sysparm_record_key=045cee1a83574f1063f6c8efeeaa3e6

Offering Corporate VPN

Open in CMDB Workspace Update Delete

Name	Corporate VPN	Approval group	
Model ID		Support group	
Owned by		Change group	
* Parent	Remote Access	Managed by	
Used for	Production	Version	
Operational status	Operational	Business criticality	2 - somewhat critical
Vendor		SLA	
		Location	
Short Description			
Description			

3. Incident Details showing category, subcategory, urgency, service, service offering and CI.

← → ↺

dev224508.service-now.com/now/sow/record/incident/8487be1283d74f1063fc8efeeaad346/params/query/active%3Dtrue%5Assigned_to%...

☆

🏠

D School

⋮

servicenow

All Favorites History ⋮

Service Operations Workspace ☆

🔍 Search

⌵

🔗

🕒

🔔

👤

🏠

☰ List INC0010001 +

⋮

📄

Unable to connect to Corporate VPN from home o... 🗒

Save

Create change request ⌵

Resolve

⋮

☰

Overview Details Related records

⋮ Incident

Short description *

Unable to connect to Corporate VPN from home office

Description

Number INC0010001

State In Progress ⌵

Stacked view

Comments More ▾

👤 Everyone can see this comment ➡

Activity 1 🔍 🔍 🔍 🔍 🔍

🔴 Beth Anglin

Field changes • 2026-09-15 08:49:42

Record Information

Last updated by Beth Anglin

2026-09-15 08:49:42

Caller

MH Michael Hoefer

Development

Brasilia - 08:49:51

America/Los_Angeles

Contact

Recent incidents >

4. Agent Assist / Knowledge article attached to the incident.

ServiceNow interface showing incident details for INC0010001. The incident title is "Michael advised he was able to connect yesterday ev...". The "Cause" field is empty. The "Resolution" field is set to "-- None --". The "Activity" section shows a comment by Beth Anglin dated 2026-09-15 09:01:54, with the text "Follow the Steps and Resolve the Incident" and a link to "KB0000003 : Managing Settings in Internet Explorer 10 for Windows 8". The "Record Information" panel on the right shows the article title, a 3-minute read time, and an average rating of 0/5.

5. Incident reassigned to Network / Level 2.

ServiceNow interface showing incident details for INC0010001. The incident title is "Unable to connect to Corporate VPN from home o...". The "Incident" section shows the caller as "Michael Hoefer", location as "SHS quadra 5, Bloco E", channel as "Phone", and category as "Network". The "Urgency" is set to "2 - Medium" and "Priority" is "4 - Low". The "Email" section shows a message to Michael Hoefer with Beth Anglin in the "Cc" field. The "Record Information" panel shows the last update by Beth Anglin on 2026-09-15 08:50:46.

6. Level 2 incident view with Task SLA in Related Records.

ServiceNow interface showing incident details for INC00000031. The incident title is "Michael advised he was able to connect yesterday ev...". The "Summary" section shows the short description: "Michael advised he was able to connect yesterday evening, but this morning received an authentication error. The Internet works fine". The "Activity" section shows a comment by Beth Anglin dated 2026-09-15 09:01:54, with the text "Follow the Steps and Resolve the Incident" and a link to "KB0000003 : Managing Settings in Internet Explorer 10 for Windows 8". The "Record Information" panel shows the last update by David Loo on 2026-09-15 09:17:45, and the "SLAs and timings" section shows a "Response SLA" of "Completed" and a "Resolution SLA" of "1wk 0d".

7. Incident On Hold with Awaiting Change.

The screenshot shows the ServiceNow Incident form for INC0010001. The incident is titled "Michael advised he was able to connect yesterday ev...". The state is "On Hold" with the reason "Awaiting Change". The caller is Michael Hoefer. The location is SHS quadra 5, Bloco E. The impact is 3 - Low. The channel is Phone. The urgency is 2 - Medium. The activity log shows a relationship change by David Loo on 2026-09-15 09:20:14. The SLAs and timings section shows the Response SLA as Completed and the Resolution SLA as 1wk 0d. The caller information is Michael Hoefer, Development, Brasilia - 09:20:17, America/Los_Angeles.

8. Child Incident listed under the parent incident.

The screenshot shows the ServiceNow Incident form for INC0010002. The incident is titled "Unable to connect to Corporate VPN". The state is "On Hold" with the reason "Awaiting Change". The description is "User receiving VPN authentication failure error". The caller is David Loo. The activity log shows a field change by David Loo on 2026-09-15 09:24:22. The Agent Assist section shows an article titled "Eclipse configuration for Android d..." with the text "Installing the Eclipse Plugin Android offers a custom plugin for the Eclipse IDE, called Androi...".

9. Cause and Resolution details with Resolved state.

The screenshot shows the ServiceNow Incident form for INC0010002. The incident is titled "Michael advised he was able to c...". The state is "Resolved". The resolution code is "Workaround provided". The resolution notes are "Restarted VPN-SRV-02 service as per emergency change request". The activity log shows a message: "INC0010001 and its child incident(s) have been resolved". The SLAs and timings section shows the Response SLA as Completed and the Resolution SLA as Paused. The caller information is Michael Hoefer, Development, Brasilia - 09:27:55, America/Los_Angeles.

10. Knowledge article created from the resolved incident.

servicenow

AllFavoritesHistory

Service Operations Workspace

Application scope: Global
Update set: Default [Global]

List

INC0000031

INC0010001

KB0010001 v0.0

+

Details

INC0010002

Michael advised he was able to connect yesterday ev...

Save

Create change request

Close

Reopen

Overview

Details

Related records

Task SLAs (3)

Affected CIs (1)

Impacted Services/CIs (1)

Child Incidents (1)

Attached Knowledge (1)

Change Requests (0)

Outages (0)

Attached Knowledge

Last refreshed 4m ago.

1 condition

Sort by

Group by

	Knowledge article	Short description	Knowledge base	Publishe
☐	KB0000003 v1.0	Managing Settings in Internet Explorer 10 for Windows 8	IT	2026-06

Contact

Recent incidents >

Recent interactions >

Assigned assets >

Assigned to

David Loo

Reassign

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Limitations
Automatic priority calculation reduces manual selection.	Behavior depends on correct ServiceNow configuration.
Controlled classification improves consistency.	Production performance depends on the ServiceNow environment.
Assignment groups provide clear ownership.	Roles/permissions may vary by instance.
Lifecycle controls improve traceability.	Enterprise SLA/integration behavior is outside the demonstrated scope.
Platform Analytics provides operational visibility.	Performance values require execution in the target instance.
Knowledge/change linkage provides supporting context.	Project is primarily a ServiceNow configuration/automation implementation.

9. CONCLUSION

Incident Lifecycle Automation in ServiceNow provides a structured solution for managing IT incidents from creation through resolution and closure. The project combines ServiceNow Incident Management, classification, Impact/Urgency based priority automation, assignment groups, lifecycle controls, resolution handling, related records and Platform Analytics.

The implementation demonstrates how ServiceNow can be configured to reduce repetitive manual work while improving consistency and visibility. It also provides a clear basis for functional and performance-oriented testing.

10. FUTURE SCOPE

- Integrate additional notification and communication channels.
- Introduce richer SLA monitoring and escalation rules.
- Extend automated assignment using additional incident attributes.
- Add advanced analytics and trend-based reporting.
- Integrate CMDB and configuration-item information more deeply.
- Introduce automated knowledge suggestions for recurring incidents.
- Extend workflow automation and enterprise integrations.

11. APPENDIX

Source Code (if any)

The project uses ServiceNow configuration and JavaScript-based Client Script logic. Project-specific configuration is maintained inside the ServiceNow instance.

Dataset Link

Not applicable. This is a ServiceNow ITSM workflow/configuration project and does not depend on an external machine-learning dataset.

GitHub & Project Demo Link

Github : <https://github.com/DLBhavana/Incident-Lifecycle-Automation>

Project Demo :

https://drive.google.com/file/d/1ljEM2iCtERVIRsYAH_zpnbAYMuJ4Wlz/view?usp=sharing

Reference Format

This report follows the supplied Final Report Template, including the specified sections and appendix items.

END OF REPORT